FullAUC Optimization for Open-Set Recognition in Remote Sensing Images

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***Abstract*—To classify remote sensing images in the wild, open-set recognition (OSR) is an important solution. It aims at classifying known classes of images and identifying the unknown classes that are not seen in the training stage. It needs to reduce the empirical and open space risk. The most prevalent prototype-based methods, which use very few prototypes to represent a class of images, may not fully capture the diversity of the images and hence increase open space risk. To address this issue, in this article, we propose an OSR method, named full area under the receiver-operating-characteristic curve (FullAUC), and applied it to remote sensing image classification. FullAUC minimizes both the risks in terms of AUC. The empirical risk is minimized by a multiclass AUC optimization algorithm on known classes. The open space risk is minimized by a binary-class AUC optimization, which discriminates all known classes against unknown classes. The advantage of FullAUC lies in that, by leveraging AUC-based ranking loss on data-to-data relations, FullAUC better captures data diversity than prototype-based methods. Another novel contribution is that we applied the background classes as a novel regularization strategy to remote sensing image classification, where background classes have been proven to be helpful in improving the generalization ability in the general OSR studies. Finally, to further enhance the detection of unknown classes, we employ a joint confidence-based decision rule with a sigmoid output layer, which mitigates the closed-set limitations of softmax. Experiments on several remote sensing benchmark datasets demonstrate the effectiveness of the proposed method. The source code is available at https://github.com/zjyellow/FullAUC**

***Index Terms*—AUC optimization, background class regularization, open-set recognition (OSR), remote sensing image classification.**

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I. Introduction

With the continuous advancement of Earth observation technologies, spaceborne and airborne platforms have generated massive volumes of remote sensing images, enabling a wide range of applications such as urban planning, environmental monitoring, and natural resource management [1]. Despite the remarkable progress achieved by deep learning-based algorithms in remote sensing image classification [2], [3], [4], [5], the field of remote sensing image interpretation still faces several inherent challenges.

One of the challenges in remote sensing is that the high resolution of hyperspectral images is characterized by high dimensionality with limited labeled data [6], which significantly increases the difficulty of model training and generalization. Another challenging issue is that the real-world scenarios of remote sensing often contain complex backgrounds, small objects, and large intraclass variations across multiple scales, which cause ineffective feature extraction and low classification accuracy [7], [8], [9], [10], [11]. In addition to these challenges, real-world open-set problems in remote sensing bring additional difficulty, where an algorithm may often encounter a scenario where a test image belongs to an unknown category. Traditional algorithms [12], [13], which mainly use samples from known classes to learn the feature space, ignore the distribution of the samples from unknown classes. Eventually, they may wrongly classify samples from the unknown classes into one of the known classes with overconfidence.

To address this challenging problem, open-set recognition (OSR) techniques [2] have been studied in remote sensing image classification. The representative OSR methods in remote sensing can be divided into softmax-based methods and prototype-based methods. da Silva et al. [2] used a threshold after softmax outputs for unknown detection in remote sensing. Gawlikowski et al. [14] applied a Dirichlet prior network to quantify the distributional uncertainty of remote sensing models. Liu et al. [15] combined the classification task with the reconstruction task to identify unknown classes. However, their performance is limited due to the closed-set property of softmax. Recent works [16], [17], [18] show that the prototype-based methods can improve the class representation of remote sensing models. For example, Du et al. [16] proposed a prototype network for the OSR in remote sensing, which utilizes the similarity between instances and prototypes for classification. Fu et al. [18] fused global and local features

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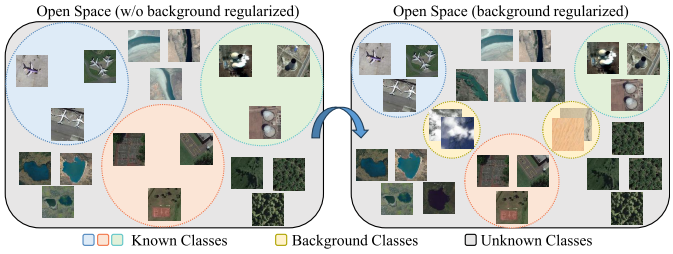


Fig. 1. Illustration of class types of OSR with/without background classes. The rounded circles include known classes and background classes of the closed space. The rest denotes unknown classes of the open space. The closed space is more compact with background class regularization (right) than without one (left).

to improve the prototype representation. Brilhador et al. [17] proposed to learn an embedding space by a prototype network that is optimized by metric learning. However, these methods are constrained by a strict data selection strategy for prototype learning. Furthermore, compressing intraclass features into very few points may lose essential discriminative information about unknown classes [19].

Although the OSR problem has been studied sufficiently in the remote sensing image classification, one crucial aspect that has been largely overlooked is the background class regularization [20]. Specifically, in the general OSR studies, there are three types of training data.

1) *Known Classes:* Important classes with manual labels in the training stage.
2) *Background Classes:* Irrelevant classes that can be obtained in the training stage. They have no overlap with the known classes.
3) *Unknown Classes:* Classes that do not appear in the training stage and may exist in the test stage.

As shown in Fig. 1, background classes introduce additional structure into the feature space, which enhances the distinction between the known and unknown classes. Without the background classes as shown in Fig. 1 (left), the feature space tends to be loosely structured, increasing the risk of misclassification for unknown classes. In contrast, with the background class regularization as shown in Fig. 1 (right), the feature space becomes more compact. It helps the model recognize unknown classes with greater confidence. Recent OSR studies have demonstrated that incorporating background classes can mitigate network agnostophobia [20], such as the study in the generic image datasets like CIFAR-10 [21]. However, existing OSR studies in the specific remote sensing field primarily focus on distinguishing between known and unknown classes while ignoring the potential benefits of background classes and, hence, their effectiveness on OSR in remote sensing remains unverified.

To address the above problems, we propose a novel full area under the receiver-operating-characteristic curve (FullAUC) optimization algorithm for OSR in remote sensing images. Specifically, we propose a multiclass-AUC optimization term to reduce the *empirical risk* on known classes and propose a binary-class-AUC-based regularization strategy with background classes to reduce the *open space risk* on unknown classes. In the test stage, in order to determine a key threshold to distinguish known and unknown classes, we employ a joint confidence-based rule for unknown-class detection, which employs a sigmoid output layer instead of the conventional softmax output layer, so as to avoid the closed-set classification property of the latter. This formulation builds high-quality decision boundaries between known and unknown classes for further reducing the open space risk.

The novelties of this article are summarized as follows.

1) *A full-AUC-based bipartite ranking algorithm was proposed for OSR and applied to remote sensing image classification.* Unlike existing prototype-based methods, which explore the prototype-to-data relationship, it minimizes the structural risk over the pairwise data-to-data relationship in the entire open space.
2) *Background classes were proposed as a novel regularization for the OSR problem of remote sensing.* The binary-class AUC term in the FullAUC treats all known classes as a positive set and background classes as a negative set in the binary-class AUC optimization. The utilization of background classes improves the robustness of unknown-class detection in the test stage.

We conducted extensive experiments on remote sensing images. The results show that the proposed method achieves the state-of-the-art performance on the OSR tasks both with and without background classes, which effectively mitigates the negative impact of unknown background classes and enhances the model's generalization capability.

The remainder of this article is organized as follows. Section II reviews some related work. Section III presents preliminaries and notations. Section IV presents the proposed method, including the network architecture, optimization objectives, optimization framework, and unknown-class detection method. Section V describes the experimental results. Finally, we summarize this article in Section VI.

II. Related Work

In this section, we first provide a brief review of remote sensing image classification and OSR-related literature. Then, we review some closely related topics, including the AUC maximization, background class regularization, and out-of-distribution (OOD) detection.

A. Remote Sensing Image Classification

In recent years, the deep learning-based methods have been applied to remote sensing images [12], [13]. These studies have achieved promising performance on the closed-set assumption. However, in real-world scenarios, unknown classes are always present; these methods may fail by wrongly classifying unknown classes as any known class. In recent years, the open-set problem in remote sensing scenarios has received attention [22], [18]. The prototype-based methods have become one of the state-of-the-art methods in open-set remote sensing classification [15], [16], [17], [23], [24]. Liu et al. [15] proposed a multitask network (MDL4OW), consisting of a classification task for known-class classification and a reconstruction task for unknown-class detection.

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Brilhador et al. [17] proposed a prototype-based method with metric learning (PTL) for detection. However, these prototype-based methods rely on strict data selection strategies and are challenging to apply in real-world scenarios.

In this article, we introduce background classes for OSR in remote sensing images, using FullAUC objective with a one-vs-rest network (OVRN) to learning the data representation, avoiding the limitations of prototypical network.

B. Open-Set Recognition

In computer vision, Scheirer et al. [25] first explored the essence of OSR and defined it as a constrained minimization problem, which motivates a series of machine learning-based studies [26], [27], [28]. Current studies can be divided into discriminative methods and generative methods.

1) Discriminative Methods: Recently, the OSR studies based on deep learning have become popular [29], [30], [31]. Bendale and Boult [32] proposed Openmax to model the distance of activation vectors for detection. Jang and Kim [33] introduced OVRN to overcome the overconfident problem. Wang et al. [34] proposed a multiexpert model (MEDAF) to reduce open space risk. Particularly, prototype-based methods, which demonstrated significant performance improvement in the open-set settings, have received much attention. Yang et al. [35], [36] proposed convolutional prototype learning (GCPL), which replaces the closed-set nature of softmax with a convolutional prototype network tailored for the OSR. Chen et al. [37], [38] proposed reciprocal point learning (RPL) and its extension version (ARPL), which discriminates samples based on the differences between the samples and the reciprocal points. The aforementioned methods achieved state-of-the-art performance in OSR on generic image datasets. However, they explored the relationship between prototypes and data only, which suffers from information loss of data and hence does not maximize the discriminability of the learned representation of data.

2) Generative Methods: The early works consider using generative methods to generate fake samples as unknown classes during training. Ge et al. [39] and Neal et al. [40] used generative adversarial networks to generate counterfactual samples for modeling unknown classes. Some studies [41], [42], [43] have shown that manifold mixup [44] outperforms GAN-based methods. However, most generative methods were designed for in-distribution data and struggle to generate truly OOD samples.

3) Backgound-Class Regularization: To our knowledge, the majority of the OSR studies did not consider the background classes that are easily obtainable in real-world scenarios. The research on semi-supervised learning suggests that background classes contain useful information that can help improve classification accuracy [45]. Then, Dhamija et al. [20] and Cho and Choo [46] both introduced background classes into the OSR study, where they proposed to combine the softmax loss on known classes and a novel loss on background classes. However, all of the aforementioned methods formulate the OSR problem with the background class as a $(K+1)$-class classification problem. This inevitably disrupts the inherent distributions of unknown classes and incurs imbalances between known and unknown classes [47].

Our method is also based on deep learning. However, unlike existing methods, the proposed method optimizes a FullAUC objective with an OVRN as the backbone. Moreover, the proposed method learns the data representation from the data-to-data relationship, which addresses the limitation of the prototype-based methods. And different from the above $(K+1)$-class methods, we propose a novel K-class method without using an explicit class to incorporate the background data during training. It essentially uses the background classes to learn an adaptive threshold for detecting open-set samples.

C. AUC Maximization

AUC is an important evaluation metric in various real-world tasks [48]. It overcomes the weakness of classification accuracy on class-imbalanced data. For example, Cortes and Mohri et al. [49] proved that maximizing classification accuracy cannot effectively maximize the AUC score. Therefore, a good thought is to maximize AUC directly. Because AUC is nonconvex and discrete, directly optimizing AUC typically leads to an NP-hard problem. As a result, a set of surrogate loss functions were proposed to reduce the computational complexity. Yan et al. [50] used a differentiable sigmoid function approximation instead of the 0–1 objective function, which makes the gradient-based methods applicable to the training of the classifiers.

In recent years, the AUC maximization based on deep learning has been successfully applied to address the class-imbalanced problems in various fields [51], [52], [53]. Wang et al. [41] proposed an AUC-maximization-based method (OpenAUC) to address the OSR problem. However, they only considered maximizing the AUC between known and unknown classes by generating fake samples to solve the open space risk, while they used a traditional softmax-based method to minimize the empirical risk, which not only did not uniformly solve the OSR problem but also increased the cost of the method.

Compared to the above methods, our method optimizes AUC without generating samples. Moreover, different from the above methods that focus on the AUC optimization problem only, our method formulates the empirical risk minimization component of OSR as a multiclass AUC maximization problem and formulates the open space risk minimization component as a binary AUC maximization problem, which uniformly solves the OSR problem only by the AUC maximization.

D. OOD Detection

OSR is closely related to the task of OOD detection [54]. Particularly, OOD detection focuses specifically on differentiating OOD samples from in-distribution samples during testing. The difference between OSR and OOD lies in that OOD does not need to correctly classify the in-distribution samples.

Hendrycks and Gimpel [54] first defined the task of the OOD detection formally. They demonstrated that the

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OOD samples have a lower maximum softmax probability than the in-distribution samples. Liang et al. [55] used input perturbation and temperature scaling to increase the separability of the in-distribution samples and OOD samples. Liu et al. [56] replaced the softmax confidence scores with energy scores for OOD. Wei et al. [57] demonstrated that logit normalization can enable the network to produce highly discriminative confidence scores for both the in-distribution and OOD samples. Hendrycks et al. [58] observed that utilizing an auxiliary set of OOD samples can enhance the model's ability to differentiate the in-distribution and OOD samples. Some studies [59], [60] used generative methods to produce OOD samples. Huang et al. [61] utilize gradient information to detect OOD inputs. Hendrycks et al. [62] employed large-scale pretrained models to address challenging OOD tasks.

In this article, we applied the proposed method to OOD successfully in Section V-F.

III. Preliminaries

In this section, we review the calculation methods of the binary-class AUC as well as two forms of multiclass AUC.

A. Binary-Class AUC

AUC is essentially an evaluation metric for binary-class problems. Given a binary-class dataset $\mathcal{X} = \{(x_n, y_n)\}_{n=1}^{N}$, where x_n is the nth sample and $y_n \in \{0, 1\}$ is the label of x_n. Then, the output of a binary-class classifier $f(x)$ on $\mathcal{X}$ can be divided into two subsets: $\mathbb{P} = \{f(x_n), \forall x_n \in \mathcal{X} | y_n = 1\}$ and $\mathbb{N} = \{f(x_n), \forall x_n \in \mathcal{X} | y_n = 0\}$. AUC is equivalent to the Wilcoxon statistic on the two subsets, which is calculated as the possibility of the correct ranking between any pair of random samples with distinct labels

$$F_{\mathrm{AUC}}(x) = P\left[f(x_1) \geq f(x_2) \,|\, f(x_1) \in \mathbb{P}, f(x_2) \in \mathbb{N}\right] \tag{1}$$

where $f(x_1)$ and $f(x_2)$ are randomly sampled from $\mathbb{P}$ and $\mathbb{N}$, respectively, and $F_{\mathrm{AUC}}(x) = 0.5$ means tied decision. We usually switch the two subsets when the AUC score is less than 0.5.

B. OvR-Multiclass AUC

Multiclass AUC is mainly an extension of the binary-class AUC to the multiclass classification problem via the framework of the one-vs-one and one-vs-rest codes. The one-vs-rest form is as follows.

Specifically, given a K-class dataset $\mathcal{X} = \{(x_n, y_n)\}_{n=1}^{N}$ with $y_n \in \{1, \ldots, K\}$ and $K > 2$, we denote the data of each class as a set, i.e., $\mathbb{C}_k = \{\forall x_n \in \mathcal{X} | y_n = k\}$, $\forall k = 1, \ldots, K$. Suppose that the probability output of a multiclass classifier $f(x)$ is $[f_1(x), \ldots, f_K(x)]$, and then, the output also can be devided into K subsets $\mathbb{P}_k = \{f_k(x_n), \forall x_n \in \mathcal{X} | y_n = k\}, k = 1, \ldots, K$, where $f_k(x_n)$ is the probability of x_n belonging to the kth class.

OvR decomposes a K-class classification problem into K binary-class problems. Given any of the K classes in the original multiclass problem, supposed to be the kth class, OvR formulates the kth class as a "positive" set $\mathbb{P}_k$ in the new binary-class problem and takes all rest $(K-1)$ classes as a "negative" set $\mathbb{N}_k$. The AUC $F_k(x)$ on the binary-class problem is calculated by the following equation:

$$\begin{aligned} F_k(x) = {} & \mathbb{E}\left[I\left[f_k(x_1) > f_k(x_2)\right] | f_k(x_1) \in \mathbb{P}_k, f_k(x_2) \in \mathbb{N}_k\right] \\ & + \frac{1}{2}\mathbb{E}\left[I\left[f_k(x_1) = f_k(x_2)\right] | f_k(x_1) \in \mathbb{P}_k, f_k(x_2) \in \mathbb{N}_k\right] \end{aligned} \tag{2}$$

where $I(\cdot)$ represents the indicator function.

Finally, the OvR-multiclass AUC is calculated as follows:

$$F_{\mathrm{OvR}}(f) = \frac{1}{K}\sum_{k=1}^{K} F_k(x). \tag{3}$$

In this article, we apply the OvR-multiclass AUC optimization with the one-vs-rest network and binary AUC optimization for background class regularization.

IV. Methodology

In this section, we present our proposed method, including the network architecture, optimization objectives, optimization framework, and unknown-class detection method.

A. Network Architecture

As shown in Fig. 2, the backbone network of the proposed method is a CNN. Its output layer is the sigmoid output layer, which is simply an OvR network with K sigmoid functions. Each sigmoid function corresponds to a binary-class AUC optimization problem on the whole remote sensing images. Using sigmoid output layer instead of the conventional softmax output layer is important for the success of the proposed method in OSR. The reason is as follows.

Although softmax is the most widely used output layer in the classification study, it may not be suitable for OSR. Due to the normalization operator in the Softmax function, it is naturally designed for closed-set problems where the entire sample space is allocated to known classes, without considering the presence of unknown classes. Therefore, directly applying softmax to OSR will misclassify the samples from unknown classes to their most similar known classes.

To address this issue, we use the sigmoid output layer instead of the softmax output layer [33]. For the integrity of this article, we present the idea as follows. Unlike the softmax function, the sigmoid function only normalizes a single logit output without involving other logits. If taking sigmoid as the activation function of the output layer, the confidence score $P(y = y_i|x)$ of the class y_i can be written as $P(y = y_i|x) = 1/(1 + e^{-f_i(x;\theta)})$, which is independent of the other classes. Moreover, the sigmoid output layer does not impose the constraint that the sum of the activations equals 1; therefore, given a sample from an unknown class, it can generate a low response at every activation, which is quite suitable for the thresholding-based detection methods.

B. Optimization Objective

Given a training set of N remote sensing images $\mathcal{X} = \{(x_n, y_n)\}_{n=1}^{N}$, where x_n denotes the nth image, $y_n \in \{1, \ldots, K, K+1, \ldots, K+B\}$ is the manual label of x_n, with

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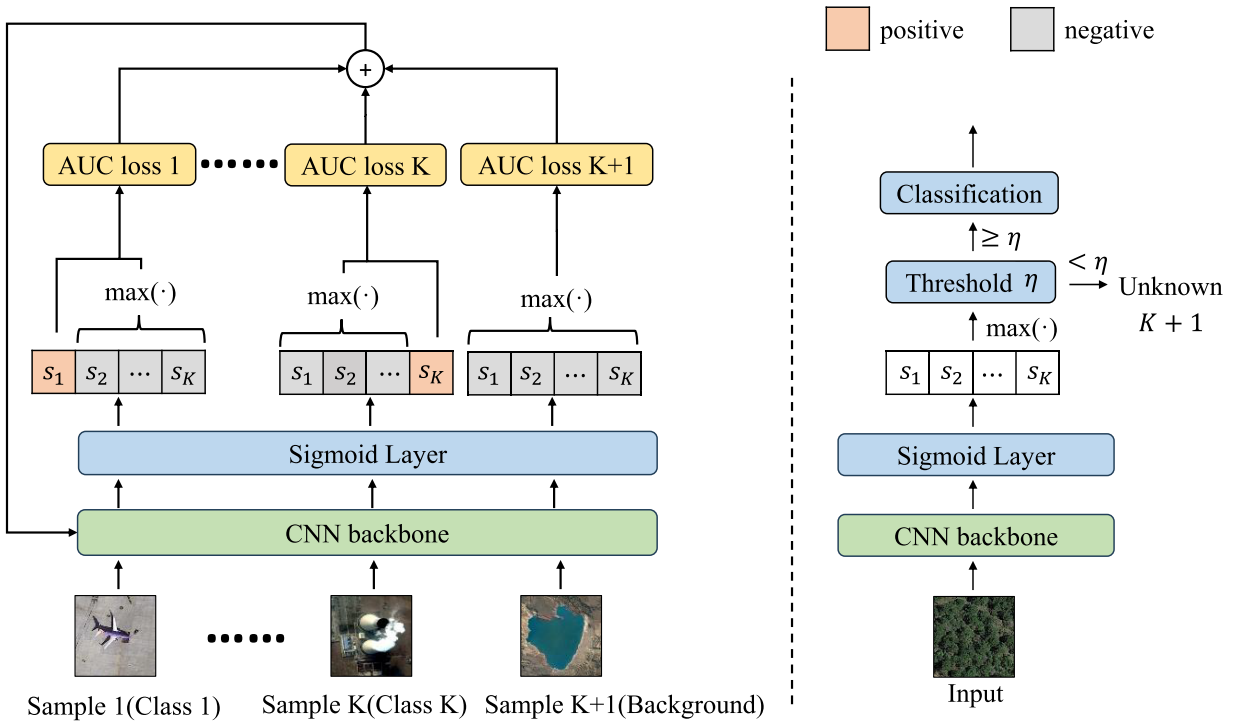


Fig. 2. Our proposed approach. (Left) Training process of the proposed method. Each sample is passed through the backbone and a sigmoid layer to obtain the probability of belonging to each class. The positive-class probability and the highest negative-class probability are fed into the AUC loss for computation. All individual losses are then aggregated to form the total loss $\mathcal{L}_{\text{total}}$ for model optimization. (Right) Inference process. The input is processed by the backbone and the sigmoid layer to produce class probabilities, and the highest probability is compared with a predefined threshold η for classification.

the first K classes as K known classes, and the latter B classes as B background classes, there is a test set of M images $\mathcal{S} = \{p_n, q_n\}_{n=1}^{M}$, where $q_n \in \{1, \ldots, K, K+1, \ldots, K+U\}$ denotes the label of the test image p_n, and U denotes the number of unknown classes.

The objective of the proposed OSR method has two components. The first component is to classify the images from the K known classes precisely, which is formulated as a problem of *minimizing the empirical risk* $\mathcal{R}_\varepsilon$ on K known classes. The other component is to detect the images that come from the U unknown classes, which is formulated as a problem of *minimizing the open-set risk* $\mathcal{R}_\mathcal{O}$ on background classes. Note that the test data from unknown categories are not needed to be classified precisely. Therefore, the objective of OSR can be written as follows:

$$\min \mathcal{R} = \min \left(\mathcal{R}_\varepsilon + \lambda \mathcal{R}_\mathcal{O}\right) \tag{4}$$

where λ is a hyperparameter for balancing the impact between $\mathcal{R}_\varepsilon$ and $\mathcal{R}_\mathcal{O}$.

The first component of the proposed method is implemented as minimizing the OvR-multiclass AUC loss on the K known classes. The second component is implemented as minimizing the binary-class AUC loss where the K known classes are integrated as the positive set and the B background classes are integrated as the negative set.

1) Closed-Set Classification Based on Multiclass AUC: Different from existing prototype-based methods, which learn prototype-to-data relationship, we aim to train a closed-set classifier on the K known classes from data-to-data relationship, which is formulated as minimizing the following OvR-multiclass AUC on the known classes:

$$\mathcal{R}_\varepsilon(x) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}\left[\ell_r\left(f_k(x_1), f_k(x_2)\right) | y_1 = k, y_2 \neq k\right] \tag{5}$$

where $\ell_r(\cdot)$ represents the 0–1 ranking loss function

$$\ell_r\left(f(x_1), f(x_2)\right) = I\left[\Delta\left(f(x_1), f(x_2)\right) < 0\right] + \frac{1}{2} I\left[\Delta\left(f(x_1), f(x_2)\right) = 0\right]. \tag{6}$$

Because the summation operator of the K binary-class AUC losses in (5) is a convex function, minimizing (5) turns into a problem of minimizing each of the K binary-class AUC losses independently. However, due to the nonconvexity and discreteness of the 0–1 ranking loss $\ell_r(\cdot)$, directly optimizing the binary-class AUC loss is an NP-hard problem [63]. We consider using a convex, continuous, and differentiable surrogate loss $\ell_s(f(x_1), f(x_2)) = f(x_1) - f(x_2)$ to solve the empirical risk minimization problem. In this work, the hinge loss is used as the surrogate loss function

$$\ell_h\left(f(x_1), f(x_2), \delta\right) = \max\left(0, \delta - f(x_1) + f(x_2)\right) \tag{7}$$

where δ is the hyperparameter to determine a margin between the output scores $f(x_1)$ and $f(x_2)$. Therefore, building on (7), the closed-set classification loss $\mathcal{L}_c$ can be represented by the following bipartite ranking function:

$$\mathcal{L}_c = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}\left[\ell_h\left(s_i^+, s_j^-, \delta\right) | s_i^+ \in \mathbb{P}, s_j^- \in \mathbb{N}_1\right] \tag{8}$$

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where $\mathbb{P}$ denotes the set of the predicted scores on the ground-truth classes; $\mathbb{N}_1$ denotes the set of the predicted scores on the wrong classes; and s_i^+ and s_j^-, respectively, are the scores in $\mathbb{P}$ and $\mathbb{N}_1$

$$\mathbb{P} = \left\{ f_j(x) \,|\, j \in \{1, \ldots, K\} \right\}, \tag{9}$$

$$\mathbb{N}_1 = \max_{1 \le i \le K} \left\{ f_i(x) \,|\, i \neq j, i \in \{1, \ldots, K\} \right\}. \tag{10}$$

It essentially enlarges the difference between the predicted scores of positive and negative sets to be larger than a predefined margin δ.

2) Open-Set Detection Based on Binary-Class AUC With Background Classes: As defined in (4), the OSR objective contains an empirical risk $\mathcal{R}_\varepsilon$ for known-class classification and an open space risk $\mathcal{R}_\mathcal{O}$ for unknown-class detection. In this section, we focus on the implementation of the open space risk.

Following the definition in [25], the open space risk measures the proportion of the open-set samples that are incorrectly recognized as known classes:

$$R_\mathcal{O}(x) = \frac{\int_\mathcal{U} g(x)\, \mathrm{d}x}{\int_\mathcal{S} g(x)\, \mathrm{d}x} \tag{11}$$

where $\mathcal{U}$ denotes the space of unknown samples, $\mathcal{S}$ denotes the total space (including known and unknown samples), and $g(x)$ is a discriminator that outputs 1 if x is classified as a known class and 0 otherwise.

To approximate and minimize $\mathcal{R}_\mathcal{O}$, we reformulate it as a binary-class AUC problem, in which all known classes form the positive set and all unknown (or background) classes form the negative set

$$R'_\mathcal{O}(x) = \mathbb{E}\left[\ell_r \left(f_k(x_1), f_k(x_2) \right) \,\middle|\, y_1 = k,\, y_2 \ge K+1 \right] \tag{12}$$

where the scores of unknown samples are expected to be lower than those of the known samples.

Since the training process only has access to closed-set data, we leverage easily obtainable background classes as a regularization strategy to enhance the robustness of the model against unknown classes [20]. The key idea is to enforce that the responses of background samples are consistently lower than those of the known samples. This is achieved via a hinge-based ranking loss

$$\mathcal{L}_{\mathrm{bg}} = \mathbb{E}\left[\ell_h \left(s_i^+, s_j^-, \delta \right) \,\middle|\, s_i^+ \in \mathbb{P},\, s_j^- \in \mathbb{N}_2 \right] \tag{13}$$

where $\mathbb{P}$ is the set of scores on the ground-truth classes and $\mathbb{N}_2$ denotes the maximum predicted scores of background class instances

$$\mathbb{N}_2 = \max_{1 \le j \le K} \left\{ f_j(x) \,\middle|\, y \ge K+1 \right\}. \tag{14}$$

This formulation effectively regularizes the model to suppress responses on background classes, which in turn reduces the open space risk.

C. Optimization Framework

Substituting (8) and (13) into (4) results in the total loss function of the proposed classifier trained with the background classes

$$\begin{aligned} \mathcal{L}_{\mathrm{total}} &= \mathcal{L}_c + \lambda \mathcal{L}_{\mathrm{bg}} \\ &= \mathbb{E}\left[\ell_h \left(s_i^+, s_j^-, \delta \right) | s_i^+ \in \mathbb{P}, s_j^- \in \mathbb{N}_1 \right] \\ &\quad + \lambda \mathbb{E}\left[\ell_h \left(s_i^+, s_j^-, \delta \right) | s_i^+ \in \mathbb{P}, s_j^- \in \mathbb{N}_2 \right]. \end{aligned} \tag{15}$$

In our work, the empirical risk and open space risk are considered to be equally significant, and both components share the same set $\mathbb{P}$. Therefore, the total loss function can be further rewritten as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathbb{E}\left[\ell_h \left(s_i^+, s_j^-, \delta \right) | s_i^+ \in \mathbb{P}, s_j^- \in \mathbb{N}_1 \cup \mathbb{N}_2 \right]. \tag{16}$$

Algorithm 1 gives the details of the proposed FullAUC optimization method.

Algorithm 1 FullAUC Optimization Algorithm

Input:
A mini-batch of input features, $\mathbf{x}$;
the corresponding labels, $\mathbf{y}$;
the number of samples in the mini-batch, N;
the margin hyper-parameter in hinge loss, δ;
initialized parameters in the convolutional layers, θ;
learning rate, φ;
the number of iterations t $\leftarrow$ 0;

Output:
Optimized parameter θ;

while the training loss does not converge **do**
  $N^+ \leftarrow \sum_{n=1}^{N} \mathbb{I}(y_n \neq K+1)$;
  $N^- \leftarrow N$;
  $t \leftarrow t+1$;
  init the positive subset $\mathbf{P}$ which contains N^+ samples, and the negative subset $\mathbf{N} = \mathbf{N}_1^- \cup \mathbf{N}_2^-$ which contains N^- samples;
  $\mathbf{p} \leftarrow f(x, \theta)$;
  for each $y_n \in \mathbf{y}$ **do**
    if $y_n \neq K+1$ **then**
      add $\mathbf{p}_{y_n}$ to $\mathbf{P}$;
      add maximum score in $\mathbf{P}$ except $\mathbf{p}_{y_n}$ to $\mathbf{N}_1^-$;
    else
      add maximum score in $\mathbf{P}$ to $\mathbf{N}_2^-$;
    compute loss $\ell^t = \frac{1}{N^+ N^-} \sum_{i=1}^{N^+} \sum_{j=1}^{N^-} \phi(s_i^+, s_j^-, \delta)$;
  compute back propagation error $\frac{\partial \ell^t}{\partial x^t}$;
  update the parameters θ by $\theta^{t+1} = \theta^t - \varphi \cdot \frac{\partial \ell^t}{\partial \theta^t}$;
return θ^*

D. Thresholding for Unknown Class Detection

During inference, unknown-class detection is performed using a thresholding strategy on the sigmoid outputs of the network, as illustrated in Fig. 2. For a test sample x, we compute the confidence scores for the K known classes

$$P(y = y_i | x) = \frac{1}{1 + \exp\left(-f_i(x; \theta)\right)}, \quad i = 1, \ldots, K. \tag{17}$$

The maximum score among the K outputs is compared with a threshold τ

$$\hat{y} = \begin{cases} \arg\max_i P(y = y_i | x), & \text{if } \max_i P(y = y_i | x) \ge \eta \\ K+1, & \text{otherwise.} \end{cases} \tag{18}$$

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This allows high-confidence samples to be classified into known classes, while low-confidence samples are classified as unknowns.

To select an effective threshold, we adopt a joint decision rule based on a development set $\mathcal{X}_{\text{dev}} = \{(m_i, n_i)\}_{i=1}^{N'}$ that only contains known-class samples. Let $f_{n_i}(m_i)$ denote the confidence score of m_i on its ground-truth class n_i. The threshold is computed as the average ground-truth confidence minus a margin δ

$$\eta = \frac{1}{N'} \sum_{i=1}^{N'} f_{n_i}(m_i) - \delta. \tag{19}$$

This rule leverages the margin constraint $s^+ - s^- > \delta$ from 13, ensuring that known-class samples lie above η, while unknown-class samples, whose activations are generally lower, fall below it. Importantly, the threshold is computed using only known-class data, which makes the approach practical in open-set scenarios.

1) Connection to OpenAUC Loss: The OpenAUC loss [41] aims to optimize the curve of a modified open-set classification rate

$$\mathcal{L}_{\text{OpenAUC}} = \mathcal{L}_C + \frac{\lambda}{|P||F|} \sum_{i}^{|P|} \sum_{j}^{|F|} \left(s_j - s_i\right)^2 \tag{20}$$

where s_i denotes the scores of a real image set P, s_j denotes the scores of a generated image set representing unknown classes, and λ denotes a hyperparameter. It contains two terms. The first term $\mathcal{L}_C$ uses the multiclass cross-entropy loss for minimizing the empirical risk on the known classes. The other term is a binary-class AUC loss for minimizing the open-set risk between the known classes and unknown classes, where it uses manifold mixup to synthesize an artificial negative set $\mathbb{F}$.

The proposed FullAUC also contains two terms. The differences between OpenAUC and FullAUC lie in that: 1) the first term $\mathcal{L}_c$ of FullAUC, which minimizes the empirical loss, is an OvR multiclass AUC loss; and 2) although the second term $\mathcal{L}_{\text{bg}}$ of FullAUC, which optimizes a binary-class AUC, is the same with OpenAUC, FullAUC introduces background classes as the negative set $\mathbb{N}$, which may learn more diverse information than the way of synthesizing a background dataset from existing data.

V. Experiments

In this section, we first introduce the datasets used in our experiments, the comparison methods, and the evaluation metrics. Then, we analyze the experimental results under the same settings. Finally, we conduct ablation studies to evaluate the contributions of different modules.

A. Datasets

We consider the conventional setting, where the model is trained using known classes and background classes, and unknown test classes are used for detection. Our experiments are conducted on the following five benchmark remote sensing image classification datasets.

1) *NWPU-RESISC45 [7]:* It consists of 256 × 256 pixeled images from 45 scene classes, each of which contains 00 images. We adopted ten classes as known classes, 25 classes as background classes, and the remaining ten classes as unknown classes for testing.
2) *RSSCN7 [8]:* It consists of 400 × 400 pixeled images from 7 scene classes, each of which contains 400 images. We adopted three classes as known classes, two classes as background classes, and the remaining two classes as unknown classes for testing.
3) *EuroSAT [9]:* It consists of 64 × 64 pixeled images with from ten scene classes, each of which contains between 2000 and 3000 images. We adopted three classes as known classes, four classes as background classes, and the remaining three classes as unknown classes for testing.
4) *SIRI-WHU [10]:* It consists of 200 × 200 pixeled images from 10 scene classes, each of which contains 200 images. We adopted four classes as known classes, four classes as background classes, and the remaining two classes as unknown classes for testing.
5) *AID [11]:* It consists of 600 × 600 pixeled images from 30 scene classes, each of which contains between 220 and 420 images. We adopted ten classes as known classes, ten classes as background classes, and the remaining ten classes as unknown classes for testing. The input samples from the above datasets are in RGB types and are resized to 64 × 64 size. We used random cropping and random horizontal flips to augment the data.

B. Comparison Methods

We compare the proposed method with 11 common and state-of-the-art comparison methods in both OSR and remote sensing studies, which can be categorized into: 1) the softmax-based methods, including the Softmax and OpenAUC [41], and MEDAF [34]; 2) the prototype-based method GCPL [36], CAC [64], MDL4OW [15], and PTL [17]; 3) the reciprocal-point-learning-based methods, including RPL [37] and ARPL [38]; and 4) the OvR network-based method, OVRN [33] and Macro-AUC [65], with our modified version of Macro-AUC to adapt it to the OvR setting. Note that the proposed FullAUC belongs to the last category. All the baselines were reimplemented with the same network in our experiments.

Most of the competitive methods did not use background classes in their original publications. To compare fairly with the competitive methods, we adopted a common method on utilizing additional training data, which is to add a **filler class** besides the known classes at the output layer of the backbone network to describe unknown classes. During the model training of the competitive methods, the background classes behave as the filler class. For method MDL4OW, the extreme value theory-based decision was used for unknown detection.

We adopted VGG32 as the backbone. We trained the network for 600 epochs, with Adam as the optimizer, and a batch size of 32 on each dataset. The initial learning rate of the network starts from 1×10^{-4} for the proposed FullAUC.

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We used the cosine annealed schedule [29] to control the learning rate resetting it to the initial rate at the 200th and 400th epochs. This schedule helps the models get rid of bad local minima. More details were listed in our code.[1] We make five independently random trials for each method and report the average results.

C. Evaluation Metrics

We used the most commonly used OSR evaluation metrics, which are described as follows.

Closed-Set Accuracy (ACC): ACC evaluates the closed-set classification performance on known classes.

Area Under Open-Set ROC Curve (AUROC): AUROC evaluates the open-set AUC performance of a model on discriminating known classes from unknown classes.

Open-Set Classification Rate (OSCR): OSCR is an ROC curve adapted from the detection and identification rate (DIR) curve [66]. Specifically, given a detection threshold η for discriminating known classes from unknown classes, OSCR uses the correct classification rate (CCR) to describe the fraction of the samples of the known classes whose probability of their ground-truth classes is maximal and greater than η

$$\mathrm{CCR}(\eta) = \frac{|\{x \in \mathcal{X} \wedge P(\hat{y}|x) \geq \eta \wedge \arg\max_y P(y|x) = \hat{y}\}|}{N}.$$

OSCR uses the false positive rate (FPR) to describe the fraction of the samples of the unknown classes whose probability of belonging to any known classes is greater than δ

$$\mathrm{FPR}(\eta) = \frac{|\{x|x \in \mathcal{S} \wedge \max_y P(y|x) \geq \eta\}|}{M}.$$

By varying the threshold η, we can draw an OSCR-based ROC curve. To get a threshold-independent metric, we use the area under the OSCR-based ROC curve, which is also abbreviated as OSCR bravely without confusion.OSCR is an overall evaluation metric that takes the classification performance for known classes and detection performance for unknown classes into account together. We report the average evaluation scores over five independent runs.

Total Accuracy ($\mathrm{ACC_{Total}}$): $\mathrm{ACC_{Total}}$ generalizes the classification accuracy from the closed-set scenario to the open-set scenario, where the unknown classes are taken as an additional class for the ACC calculation.

Macro-F1 ($\mathrm{F1_{Macro}}$): $\mathrm{F1_{Macro}}$ is the average of all the per-class $F1$ scores

$$\mathrm{F1_{Macro}} = 2 \times \frac{\hat{P} \times \hat{R}}{\hat{P} + \hat{R}}$$

where $\hat{P} = (1/K+1)\sum_{i=1}^{K+1}(\mathrm{TP}_i/\mathrm{TP}_i + \mathrm{FP}_i)$ and $\hat{R} = (1/K+1)\sum_{i=1}^{K+1}(\mathrm{TP}_i/\mathrm{TP}_i+\mathrm{FN}_i)$ denote the averaged precision and recall values respectively; TP is the number of samples correctly classified as positive ones; FP denotes the number of samples wrongly classified as positive ones; and FN denotes the number of samples wrongly classified as negative ones.

D. Results

As shown in Table I, FullAUC achieves the highest ACC scores across all five benchmark datasets, significantly outperforming existing approaches. For instance, it reaches 98.83% on RSSCN7 and 99.80% on EuroSAT, demonstrating strong discriminative power on known classes. These results indicate that FullAUC effectively learns robust representations under the closed-set setting.

Table II presents the AUROC results for the open-set detection task. FullAUC consistently outperforms prior methods, especially on challenging datasets such as EuroSAT (96.62%) and RSSCN7 (86.03%). These improvements suggest that FullAUC is more effective at distinguishing unknown samples from known categories, which is essential for reliable open-set recognition.

The OSCR results in Table III further validate the superiority of FullAUC in realistic OSR scenarios. It achieves the highest OSCR scores on all datasets, with large margins over strong baselines such as MEDAF and OVRN. Notably, FullAUC reaches 96.22% on EuroSAT and 82.46% on RSSCN7, demonstrating its robustness in handling both known and unknown categories simultaneously.

Table IV reports the overall performance of each method when jointly evaluating closed-set classification and open-set detection using two metrics: ACCTotal and F1Macro. FullAUC significantly outperforms all baseline methods across all datasets in both metrics. Notably, it achieves 96.83% ACC and 0.9678 F1 on EuroSAT, and 90.83% ACC and 0.9075 F1 on AID, surpassing the best baseline (e.g., Macro-AUC and OpenAUC) by large margins. Compared to the second-best

TABLE I
ACC SCORES (%) OF COMPARISON METHODS ON THE CLOSED-SET CLASSIFICATION SUBPROBLEM OF OSR

Method	NWPU	RSSCN7	EuroSAT	SIRI	AID
Softmax	92.63	94.50	98.33	95.51	85.91
GCPL [37]	93.31	95.17	98.47	95.30	86.27
RPL [38]	92.91	95.50	97.23	95.51	86.18
ARPL [39]	93.20	94.00	98.20	95.51	85.45
CAC [65]	93.86	94.00	98.43	94.50	86.73
OVRN [34]	94.29	94.67	98.60	97.00	88.82
OpenAUC [42]	95.23	96.67	98.97	97.26	91.18
MDL4OW [15]	92.86	94.67	98.10	94.50	85.82
PTL [17]	93.61	96.00	97.20	96.00	83.45
MEDAF [35]	94.57	97.07	99.05	97.04	91.63
Macro-AUC [66]	94.64	98.31	99.23	95.97	91.75
FullAUC(Ours)	**95.77**	**98.83**	**99.80**	**97.75**	**92.18**

TABLE II
AUROC (%) ON THE OPEN-SET DETECTION SUBPROBLEM OF OSR

Method	NWPU	RSSCN7	EuroSAT	SIRI	AID
Softmax	84.26	74.70	86.75	75.51	80.26
GCPL [37]	85.64	73.83	88.70	79.15	82.04
RPL [38]	83.83	73.24	83.31	77.11	80.05
ARPL [39]	85.54	76.56	88.60	78.72	81.71
CAC [65]	86.48	72.45	89.60	74.96	81.48
OVRN [34]	86.01	76.87	90.49	81.76	83.86
OpenAUC [42]	82.45	77.17	88.72	80.11	82.51
MDL4OW [15]	83.48	75.53	86.89	81.25	80.02
PTL [17]	83.93	77.50	81.75	83.05	77.67
MEDAF [35]	85.66	79.10	90.39	82.63	82.97
Macro-AUC [66]	86.24	81.36	91.53	83.79	81.30
FullAUC (Ours)	**87.81**	**86.03**	**96.62**	**86.09**	**86.86**

[1] https://github.com/zjyellow/FullAUC

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TABLE III
OSCR SCORES (%) OF THE COMPARISON METHODS ON OSR

Method	NWPU	RSSCN7	EuroSAT	SIRI	AID
Softmax	79.95	70.29	86.09	72.64	69.98
GCPL [37]	81.45	71.24	88.06	74.74	73.34
RPL [38]	79.53	70.89	81.86	73.34	69.64
ARPL [39]	81.09	70.66	87.93	74.71	71.22
CAC [65]	82.26	66.64	88.87	71.77	71.67
OVRN [34]	82.26	72.09	89.97	78.28	75.46
OpenAUC [42]	79.82	74.11	87.92	76.66	75.48
MDL4OW [15]	80.09	69.02	86.78	75.47	69.36
PTL [17]	79.31	72.05	78.19	77.71	67.99
MEDAF [35]	82.04	77.28	91.74	78.35	76.26
Macro-AUC [66]	81.34	76.65	90.79	77.62	77.52
FullAUC (Ours)	**84.61**	**82.46**	**96.22**	**83.01**	**80.62**

scores, FullAUC improves ACCTotal by up to 4% and F1Macro by up to 5%. These results demonstrate that FullAUC achieves a more balanced and effective performance in handling both known and unknown categories simultaneously, making it well-suited for practical open-world scenarios.

The above results demonstrate the comprehensive advantages of FullAUC across all key aspects of open-set recognition. From closed-set classification to open-set detection and joint evaluation, FullAUC consistently achieves state-of-the-art performance across all tasks. These findings highlight its strong generalization ability, robust decision boundaries, and practical value in open-world recognition tasks.

E. Ablation Study

1) Effect of the Background Classes: As shown in Table V, our proposed FullAUC method consistently outperforms strong baselines such as ARPL, OVRN, and OpenAUC across all three metrics: ACC, AUROC, and OSCR. Without background classes, FullAUC achieves 96.29% ACC, 83.46% AUROC, and 80.10% OSCR—already surpassing all competing methods. Unlike prior works such as ARPL, which rely on reciprocal-point-based metric learning, or OpenAUC, which separately optimizes empirical risk using softmax and open space risk using AUC with synthetic negatives, our FullAUC method formulates both components within a unified AUC optimization framework. This pairwise AUC-based formulation allows FullAUC to consistently optimize ranking quality across both known and unknown categories, bridging the gap left by methods that either rely on point-based classification objectives or fixed prototypes.

As shown in Fig. 1, incorporating background classes helps to reshape the feature space by pushing known classes closer together and away from ambiguous regions. This leads to more compact intraclass distributions for known classes and reduces the chance of unknown samples being incorrectly classified into known categories. When background class regularization is applied, AUROC improves from 83.46% to 87.28% and OSCR increases from 80.10% to 84.18%. Background classes act as a structured negative set during training, effectively regularizing the decision boundary between known and unknown classes. The observed improvements confirm that background classes provide rich structural information that reduces open space risk and improves unknown-class separability, demonstrating their practical value in remote sensing OSR.

Table VI compares the performance of the proposed method using different types of background data: real data, data generated by a generative model [67], data produced via mixup [68], and combinations of real data with synthetic data. The results show that combining real data with generative model data achieves the best overall performance, with 97.12% in $\text{ACC}_{\text{Total}}$, 0.9710 in F1_{Macro}, and 96.30% in OSCR, indicating that synthetic data can effectively complement real background data and improve feature-space regularization. In contrast, using only generative models or mixup data slightly degrades performance, and combining real data with mixup data does not bring noticeable improvements, suggesting that mixup-generated samples lack sufficient diversity and discriminability. Overall, these results demonstrate that augmenting real background data with generative model samples effectively enhances open-set recognition.

2) Effect of the Filler Class: Table VII lists the performance comparison of FullAUC with Filler (F) and NF. The nonfiller (NF) of FullAUC outperforms the filler class (F) across all evaluation metrics. Specifically, NF achieves higher values in ACC, AUROC, and OSCR, indicating better open-set recognition capability. In terms of overall performance, NF also shows superior $\text{ACC}_{\text{Total}}$ (92.70%) and F1_{Macro} (0.9263), suggesting better classification accuracy and balance among all classes.

The key difference between the two settings lies in the threshold rule; the F does not use this mechanism, while the NF integrates a learned threshold to separate known and unknown samples. The results demonstrate that applying the thresholding strategy improves the model's ability to handle unknown classes and overall performance.

3) Analysis on the Hinge Loss: Table VIII compares different variants of hinge loss within the FullAUC framework. The results show that the standard hinge loss achieves the best overall performance, with the highest $\text{ACC}_{\text{Total}}$ (96.83%) and F1_{Macro} (0.9678), indicating its effectiveness in maintaining both closed-set accuracy and open-set robustness. The exponential and log losses yield slightly lower results, suggesting weaker discriminability, while the squared loss leads to a significant performance drop, especially in $\text{ACC}_{\text{Total}}$ and F1_{Macro}, likely due to its sensitivity to outliers. Overall, hinge loss proves to be the most stable and effective choice for FullAUC.

4) Effect of the Hyperparameter: The hyperparameter δ is a margin that controls the discriminability between the positive samples and negative samples. To study its effect, we tuned its value from a range of $[0, 1.0]$ with a step size of 0.1 on NWPU-RESISC45, RSSCN7, EuroSAT, SIRI-WHU, and AID. The result in Fig. 3 shows that the OSR performance is improved at first along with the increase of δ, and then degrades when δ was further enlarged. The best performance is achieved on most datasets when $0 < \delta \leq 0.3$.

As shown in Fig. 4, the performance of the model is highly sensitive to the choice of the hyperparameter λ. When λ is 0.0, the closed-set accuracy ACC remains high, but the open-set metrics, including $\text{ACC}_{\text{Total}}$, F1_{Macro}, AUROC, and OSCR, are significantly degraded, indicating poor open-set discrimination capability. Increasing λ to 0.1 leads to a substantial improvement across all open-set metrics and

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TABLE IV

ACC_{Total} SCORES (%) AND $F1_{Macro}$ OF THE COMPARISON METHODS ON THE CLOSED-SET CLASSIFICATION AND OPEN-SET DETECTION SUBPROBLEMS OF OSR JOINTLY

Method	NWPU-RESISC45		RSSCN7		EuroSAT		SIRI-WHU		AID	
	ACC_{Total}	$F1_{Macro}$	ACC_{Total}	$F1_{Macro}$	ACC_{Total}	$F1_{Macro}$	ACC_{Total}	$F1_{Macro}$	ACC_{Total}	$F1_{Macro}$
Softmax	85.14	0.8293	79.50	0.7748	87.27	0.8652	80.00	0.7711	79.17	0.7768
GCPL [37]	86.44	0.8446	81.38	0.7960	88.05	0.8730	79.60	0.7616	79.75	0.7791
RPL [38]	85.27	0.8277	81.13	0.7947	87.18	0.8663	80.40	0.7746	79.50	0.7747
ARPL [39]	85.64	0.8292	79.25	0.7713	88.15	0.8751	79.80	0.7675	78.68	0.7660
CAC [65]	86.94	0.8484	78.88	0.7696	86.40	0.8574	79.60	0.7692	80.08	0.7821
OVRN [34]	86.68	0.8406	81.25	0.7980	86.80	0.8613	81.40	0.7884	81.98	0.8057
OpenAUC [42]	87.45	0.8469	82.63	0.8059	88.30	0.8749	81.60	0.7833	83.97	0.8175
MDL4OW [15]	86.66	0.8519	83.25	0.8207	91.15	0.9094	81.80	0.8100	80.50	0.7942
PTL [17]	86.85	0.8528	83.44	0.8168	88.05	0.8779	80.60	0.7831	76.86	0.7377
Macro-AUC [66]	87.35	0.8612	84.15	0.8279	89.10	0.8849	81.65	0.7962	78.01	0.7521
FullAUC(Ours)	**92.70**	**0.9263**	**90.50**	**0.9018**	**96.83**	**0.9678**	**92.20**	**0.9204**	**90.83**	**0.9075**

TABLE V

PERFORMANCE OF OPENAUC, OVRN, ARPL, AND FULLAUC ON ACC (%), AUROC (%), AND OSCR (%) UNDER W/O BC (WITHOUT BACKGROUND CLASSES) AND WITH BC (BACKGROUND CLASSES)

Method	ACC		AUROC		OSCR	
	w/o BC	with BC	w/o BC	with BC	w/o BC	with BC
ARPL [39]	93.35	92.23	79.22	82.22	74.02	78.92
OVRN [34]	92.55	94.76	77.34	83.00	71.80	79.18
OpenAUC [42]	95.06	95.46	77.37	82.99	75.79	78.28
FullAUC (Ours)	**96.29**	**95.29**	**83.46**	**87.28**	**80.10**	**84.18**

TABLE VI

COMPARISON OF DIFFERENT TYPES OF BACKGROUND DATA FOR OPEN-SET RECOGNITION ON ACC (%), AUROC (%), OSCR (%), ACC_{Total} (%), AND $F1_{Macro}$

Type of Data	ACC	ACC_{Total}	$F1_{Macro}$	AUROC	OSCR
Real	**99.80**	96.83	0.9678	96.62	96.22
Synthetic	99.60	92.53	0.9224	92.20	91.94
Mixup	**99.80**	95.20	0.9508	94.15	93.76
Real+Synthetic	99.73	**97.12**	**0.9710**	**96.70**	**96.30**
Real+Mixup	**99.80**	96.60	0.9656	95.96	95.59

TABLE VII

PERFORMANCE OF FULLAUC WITH FILLER (F) AND NF ON ACC (%), AUROC (%), OSCR (%), ACC_{Total} (%), AND $F1_{Macro}$

FullAUC	ACC	AUROC	OSCR	ACC_{Total}	$F1_{Macro}$
Filler (F)	94.83	86.08	82.61	91.74	0.9165
Non-Filler (NF)	**95.77**	**87.81**	**84.61**	**92.70**	**0.9263**

TABLE VIII

COMPARISON OF DIFFERENT HINGE LOSS ON ACC (%), AUROC (%), OSCR (%), ACC_{Total} (%), AND $F1_{Macro}$

FullAUC	Acc	ACC_{Total}	$F1_{Macro}$	AUROC	OSCR
Hinge loss	99.80	96.83	0.9678	96.62	96.22
Exponential [71]	99.77	96.78	0.9674	96.04	95.56
Squared [72]	99.57	84.25	0.8128	95.08	94.70
Log [73]	99.67	91.58	0.9107	95.61	95.90

achieves the best overall performance. A further increase to 0.2 maintains comparable results, while larger values such as 0.5 and 2.0 cause a gradual decline in all metrics, suggesting that overly large regularization weakens both closed-set and

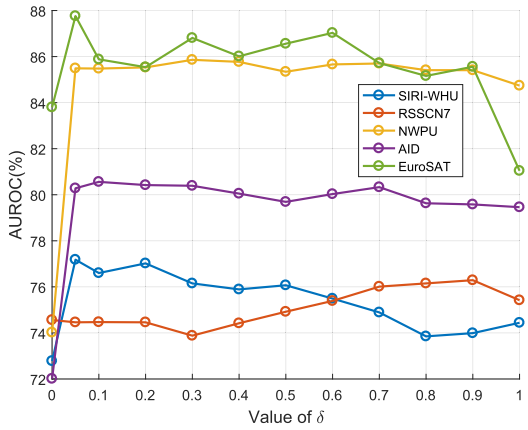


Fig. 3. Effect of the margin δ on performance.

TABLE IX

COMPARISON OF DIFFERENT UNDER ANOTHER BACKBONE LOSS ON ACC (%), AUROC (%), OSCR (%), ACC_{Total} (%) AND $F1_{Macro}$

ResNet34	Acc	ACC_{Total}	$F1_{Macro}$	AUROC	OSCR
FullAUC	**99.90**	**97.88**	**0.9786**	**97.67**	**97.36**
OpenAUC	99.80	88.80	0.8784	93.69	93.36
ARPL	99.87	90.25	0.8960	95.32	94.99

open-set performance. These results indicate that a moderate λ around 0.1–0.2 strikes the best balance between closed-set accuracy and open-set recognition ability.

5) Effect of the Backbone Networks: As shown in Table IX, we replaced the backbone with ResNet-34 and evaluated our method (FullAUC) along with other loss functions, including OpenAUC and ARPL. The results indicate that our FullAUC

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TABLE X
RESULTS OF DISTINGUISHING CIFAR10 FROM NEAR OOD DATASET CIFAR100 AND FAR OOD DATASET SVHN. ALL REPORTED RESULTS ARE IN PERCENTAGE

Method	In: RSSCN7 / Out: SIRI-WHU					In: NWPU-RESISC45 / Out: EuroSAT				
	TNR	AUROC	DTACC	AUIN	AUOUT	TNR	AUROC	DTACC	AUIN	AUOUT
Softmax	20.00	68.58	66.11	74.43	61.42	41.94	88.95	82.29	93.01	80.58
GCPL [37]	20.56	71.84	70.76	76.79	62.82	34.75	85.48	79.81	90.46	75.67
RPL [38]	27.78	71.59	70.28	79.53	64.41	33.06	85.70	78.88	91.21	74.76
ARPL [39]	28.89	75.07	70.42	79.63	69.84	34.62	86.23	82.34	92.61	70.80
CAC [65]	22.78	70.86	67.99	75.97	65.60	46.50	87.96	81.53	92.12	81.93
OVRN [34]	32.00	69.48	65.76	72.87	63.10	32.00	85.55	81.48	89.75	72.57
OpenAUC [42]	38.33	75.09	71.67	76.67	72.67	41.25	85.55	80.95	89.85	77.73
MDL4OW [15]	18.33	63.43	63.54	65.46	57.81	29.88	85.16	79.72	90.10	74.18
PTL [17]	25.00	75.48	71.74	81.66	67.80	55.56	88.42	81.75	92.19	80.32
MEDAF [35]	39.85	78.36	74.98	81.41	74.23	48.37	89.74	82.63	93.72	82.24
FullAUC (Ours)	**42.22**	**83.52**	**80.07**	**86.39**	**79.18**	**55.88**	**92.34**	**85.69**	**95.55**	**87.15**

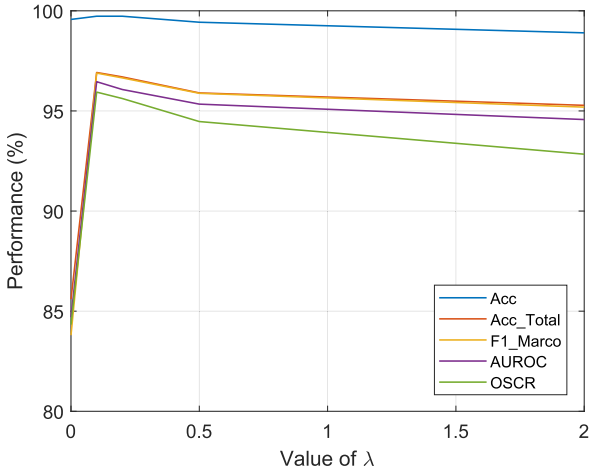


Fig. 4. Effect of the hyperparameter λ on performance.

method outperforms OpenAUC and ARPL across all evaluation metrics, demonstrating its superior effectiveness under an alternative backbone.

F. Application to OOD Detection

OOD detection is a special task of OSR. It aims to detect the samples that do not belong to any of the known classes for testing.

1) Datasets: We constructed an OOD task in two settings as follows.

In: RSSCN7, Out: SIRI-WHU: We used RSSCN7 as the in-distribution dataset for training. Then, we used SIRI-WHU as the OOD dataset.

In: NWPU-RESISC45, Out: EuroSAT: We used NWPU-RESISC45 as the in-distribution dataset for training. Then, we used EuroSAT as the OOD dataset for testing.

The overlapped classes between in-distribution and OOD data were discarded. Random cropping and random horizontal flips were used to augment all datasets.

2) Evaluation Metrics: The evaluation metrics for the OOD task include the AUROC, detection accuracy (DTACC), the area under the precision–recall curve (AUPR), and the true negative rate (TNR) at 95% true positive rate (TPR).

1) *DTACC*: It measures the maximum classification accuracy on both known and unknown classes over a range of threshold values. Note that the probability of samples from known or unknown classes is assumed to be equal.
2) *AUPR*: A precision–recall curve takes precision = TP/(TP + FP) as X-axis and recall = TP/(TP + FN) as Y-axis. It can be drawn by tuning the detection threshold. We use AUIN to denote the AUPR when in-distribution samples are classified as the positive set and use AUOUT to denote the AUPR when OOD samples are classified as the positive set.
3) *TNR at 95% TPR*: This evaluation metric reports TNR = TN/(TP + FN) when TPR = TP/(FP + FN) equals 95%.

3) Comparison Methods: We adopted ResNet34 [72] as the backbone network of all comparison methods. We trained the network for 600 epochs with a batch size of 32 and the Adam optimizer. The initial learning rate was set to 1×10^{-4}.

4) Results: Table X lists the results of comparison methods on OOD. From the table, we see that the proposed FullAUC achieves the best performance. For example, compared with MEDAF on the OOD dataset SIRI-WHU, FullAUC achieves a relative increase of 3.94% in terms of TNR, 23.84% in AUROC, 20.34% in DTACC, 26.79% in AUIN, and 19.21% in AUOUT. This demonstrates that FullAUC can effectively separate the different distributions of data.

VI. CONCLUSION

In this study, we have proposed a FullAUC optimization method for the classification problem of remote sensing images in the wild. Specifically, we proposed multiclass AUC to minimize the empirical risk on the closed-set space and employed the binary-class AUC to minimize the open-set risk, which makes the entire optimization objective an AUC-based one. The merit of adopting the AUC-based loss instead of the common classification-based loss lies in that the AUC optimization essentially minimizes a ranking loss of data, which is a structural loss that captures the correlative and discriminative information between the pairwise samples, but not merely the connection between the class centers and the pointwise samples. Then, we proposed a regularization method by leveraging background classes, which provides the

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model with sufficient and diverse knowledge of the open-set samples. Finally, to maximize the ability on discriminating the closed-set samples from the open-set samples, we employed the sigmoid output layer in the OvR network instead of the common softmax output layer and further employed a threshold-based joint decision rule instead of the common way of using a filler class for accommodating the open-set samples.

We have conducted an extensive experimental comparison in remote sensing images with the state-of-the-art OSR methods. The results demonstrate that the proposed method outperforms the comparison methods. It is also observed that the open-set performance of the proposed method is highly correlated with the closed-set performance. Finally, because the proposed method minimizes the structural loss, i.e., the ranking loss, instead of minimizing the classification errors, it works effectively on a wide OSR scenario.

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